Disclosure Regulation, Stock Liquidity, and International Policy Diffusion

Abstract

We examine stock liquidity around public-disclosure requirements and the international spread of these requirements. The policy calendar contains 211 country–family implementation milestones in 65 countries during 2000–2025. A common three-year comparison supports five families, 61 implementations and 47 countries. The equal-family stacked estimate corresponds to 17.3% lower trading costs after implementation. Market-cap alternatives, reported in the Online Appendix, are smaller and less precise. We then distinguish exposure to foreign rules, observed outcomes among earlier adopters, copying, competition and external financial dependence. Foreign-policy exposure has a more positive association with implementation when earlier adopters experience lower trading costs in the implementation-hazard models. This interaction is not confirmed by the within-country specifications or the full multiple-testing adjustment. The findings support a comparison of liquidity patterns across reporting requirements while identifying the limits of a common liquidity effect and a learning interpretation.

Keywords: disclosure regulation; stock liquidity; policy diffusion; international reporting.

1 Introduction

Governments increasingly require companies to publish information beyond their annual financial statements. Expanded auditor reports, environmental and social reports, country reports and supply-chain statements address different information needs. We ask two related questions. Do these public-disclosure requirements improve stock liquidity on average? What explains their international implementation? Figure 1 connects the source evidence to these two empirical comparisons.

Our primary country-level comparison covers 61 country–family implementations in 47 countries. Trading costs are 17.3% lower after implementation when each supported disclosure family receives equal weight. This target answers the question of an average across different types of

reporting requirements. Market-cap alternatives in the Online Appendix give smaller, less precise estimates. The comparison therefore provides evidence of lower trading costs under an equal-family target, rather than an established common benefit for all reporting requirements. The supported climate estimate concerns two selected greenhouse-gas reporting episodes, rather than every jurisdiction with climate-related reporting.

The diffusion tests relate implementation to information available from other countries. In the hazard models, the interaction between portfolio exposure to foreign requirements and observed liquidity improvements among earlier adopters is positive under both family weights. This pattern is consistent with countries responding to favourable foreign experience. It remains exploratory: the corresponding within-country interaction is not positive, and the hazard interaction does not survive the adjustment across all focal diffusion tests. We distinguish this conditional pattern from unconditional exposure, copying of larger economies, competition and financial dependence.

Disclosure research provides the basis for these questions. Information can reduce investors' trading costs by limiting differences in information and improving the precision and comparability of public reports (Diamond and Verrecchia 1991, Welker 1995, Healy et al. 1999, Leuz and Verrecchia 2000, Healy and Palepu 2001, Beyer et al. 2010). The benefits of mandatory reporting depend on reporting incentives, enforcement and the costs of producing and using information (Daske et al. 2008, Christensen et al. 2013, Christensen et al. 2016, Leuz and Wysocki 2016, Blankespoor et al. 2020). Thus, a requirement to report need not produce the same market response across countries or reporting topics.

An expanding literature examines particular reporting requirements. Research on sustainability reporting considers environmental and social information, company responses and capital-market outcomes (Christensen et al. 2021, Krueger et al. 2024). Other studies examine expanded auditor reports, electronic reporting, extractive payments and country-by-country tax information (Lennox et al. 2023, Blankespoor et al. 2014, Rauter 2020, De Simone and Olbert 2022, Müller et al. 2024, Nessa et al. 2025). These studies motivate a common comparison across reporting families. We bring the requirements into one framework while retaining their different audiences, legal scopes

and implementation dates. The analysis asks about the average pattern across supported requirements; it does not assume that all disclosure rules are equivalent.

The second question connects capital-market research to policy diffusion. Governments can learn from outcomes abroad, follow a visible example, respond to competitors or face external pressure (Simmons and Elkins 2004, Simmons et al. 2006, Shipan and Volden 2008). In accounting, Koning et al. (2018) distinguish these explanations for international IFRS adoption. We apply that logic to a broader disclosure calendar and separate foreign policy exposure from observed foreign liquidity outcomes. This distinction allows a favourable outcome abroad to enter separately from the mere existence of a foreign rule.

Our contribution is a common country-level comparison and a source-linked implementation calendar. The legal archive and the market-data universe are global; usable regression samples depend on timing, market coverage and the availability of documented comparisons. We preserve this distinction throughout. The paper first presents liquidity estimates and then examines international implementation.

2 Background

2.1 What information reaches investors?

The public-disclosure families cover financial statements under IFRS, expanded auditor reports, environmental and social reporting, climate reporting, public bank and multinational country reports, extractive payments, supply chains, sustainability-related financial reporting and electronic financial reporting. These are international families of requirements implemented through national laws, listing rules and standards. A country name identifies the jurisdiction and timing of implementation, rather than a separate country-specific research topic.

The information differs substantially. Financial-reporting reforms change the basis on which reported numbers are prepared. Expanded auditor reports explain matters such as significant audit risks and key or critical audit matters. Environmental and social requirements can mandate a report

while allowing materiality judgements or explanations for particular omitted items. We therefore distinguish a mandatory report from a mandate for every environmental, social and governance metric. Optional guidance and recommended reporting frameworks do not become mandatory simply because an exchange publishes them.

Country-by-country reporting also has distinct audiences. Public bank reports disclose specified country-level activities and financial information. Public multinational reports provide country-level tax information under separate eligibility and publication rules. Confidential tax reports are transmitted to tax authorities and are not, by that obligation alone, available to investors. Their liquidity consequences may reflect tax enforcement or other channels. We report them separately from the public-disclosure headline estimate.

More useful public information can make investors less reluctant to trade against better-informed counterparties. The expected result is lower trading costs and greater liquidity. Preparation costs, limited enforcement, boilerplate reporting, existing information and partial issuer coverage can weaken that response. A country-level estimate also includes firms outside the legal perimeter and possible spillovers. It measures a market-level response to a requirement, not an effect on every covered issuer.

Climate content cuts across reporting families. The EU Non-Financial Reporting Directive requires relevant environmental information and explicitly discusses greenhouse-gas emissions. National scope and materiality provisions matter. ESRS E1 under the CSRD subsequently specifies Scope 1, 2, 3 and total emissions, with materiality and transition provisions (European Parliament and Council 2014, European Commission 2023). Broader sustainability episodes must not be interpreted as an absence of climate reporting simply because they are not separate entries in the selected climate series. The two-country estimate is a selected episode comparison, not an exhaustive global climate-disclosure effect.

Related studies supply institutional benchmarks. Downar et al. (2021) examine UK carbon reporting, Joshi et al. (2020) examine public bank country reports, and She (2022) studies California transparency. Baik et al. (2024) studies a different US supply-chain requirement concerning

conflict minerals. Published samples do not cover identical countries, rules or years. The Online Appendix links every family to the literature and separates historical published country lists from the current source calendar.

2.2 Why might a reporting requirement spread?

Learning requires information about outcomes. A country exposed to an earlier adopter may become more willing to implement a requirement when that adopter experiences a favourable market outcome. Exposure alone does not establish learning: foreign requirements may be visible without producing an observed benefit.

Copying instead concerns the visibility or legitimacy of an example. A larger nearby economy or countries sharing a language may provide a familiar reporting template. Competition concerns the possible cost of lagging behind an economic rival. External pressure concerns institutions or financing relationships that influence domestic choices. Coordinated legal obligations, including EU requirements, differ from independent national decisions. We separate those implementations when examining the channels proposed by Koning et al. (2018).

3 Data and Research Design

3.1 Sample and datasets

The research universe contains 90 Datastream jurisdictions and market data for 2000–2025. Selected source-reviewed legal provisions are available for 81 of those jurisdictions. We retain statutes, standards, official notices, amendments and successor references, with document hashes and links between the sources and coded facts. Scanned passages used in the review are checked with OCR and page images. The archive records uncertainty rather than treating an unresolved date or scope as nonadoption.

The empirical public-disclosure calendar contains 211 dated country–family milestones in 65

countries during 2000–2025 (Table 1 and Figure 2). It records implementation phases rather than an exhaustive census of first-ever national disclosure laws. The full adoption risk panel, including the separate confidential-tax comparison, covers 82 countries; the public-reporting risk panel covers 66. Regression samples are smaller where a particular channel or market measure is unavailable.

We distinguish legal adoption, entry into force, the first covered financial period, an engagement boundary and a statutory filing deadline. The calendar preserves the clock supported by the source. For example, China's initial cross-listed audit phase differs from its later broader phase; Canada's mandatory TSX phase differs from the earlier general standard; and Dutch expanded reporting precedes the later EU reporting milestone. Vietnam's greenhouse-gas addition applies after the grandfathered 2020 annual report, whose requirements remain under the predecessor circular. The Moroccan first implementation deadline is a filing obligation, not an observed issuer receipt. We do not invent actual filing dates or require complete national histories for an otherwise supported implementation milestone. The evidence supplement lists the reconciled cases and their coverage.

The market data provide three liquidity proxies following the reporting literature (Amihud 2002, Lesmond et al. 1999, Krueger et al. 2024). *Bid-Ask Spread* measures the cost of trading immediately; lower values indicate greater liquidity. *Price Impact* measures the association between absolute returns and dollar trading volume. *Zero Returns* is the fraction of observed trading days without a price change. We use log trading costs and log price impact, screen for sufficient annual observations, and winsorize the first two measures at the annual first and ninety-ninth percentiles. An additional illiquidity index averages the three standardized measures. Higher values of each outcome indicate less liquidity.

World Bank data provide income, GDP growth, trade, aid and IMF credit measures. IMF bilateral portfolio positions measure financial exposure to foreign markets. Fixed geographic, language and export-pattern networks identify other peer groups. These public country data complement the licensed market observations; the firm-level Datastream files are not publicly redistributed.

3.2 Liquidity comparisons and weights

We form a separate comparison for each supported family and implementation cohort. For a milestone in year g, the primary window contains years $g-3$ through $g-1$ and $g+1$ through $g+3$. We omit year g as a transition year. Comparison countries have a documented later milestone after the outcome window. An unknown date does not supply an untreated observation. This construction avoids using already implemented cases as later-adopter controls for the same series, while recognising that countries may have other or earlier reporting requirements (Goodman-Bacon 2021, Callaway and Sant'Anna 2021, Sun and Abraham 2021, Baker et al. 2022).

We choose each company's largest positive-market-value security line in year $g-1$ and hold the companies fixed over the window. A country must supply at least 30 qualifying companies with the required annual market measures. This gives a comparison among firms with persistent observable trading data and limits changes in company composition. It also reduces country coverage and can favour larger, more actively traded companies. Two-year windows assess this restriction with a different horizon; they do not provide an identical sample.

Within each cohort we estimate a country- and year-effect comparison, with country weights based on GDP in $g-1$. Cohort weights within a family reflect the baseline GDP represented by adopting countries. The primary sample contains 1,404 country–stack–years, 47 countries, 61 country–family implementations and 18 family–cohorts. Country-cluster scores retain dependence when a country appears in several cohorts or families.

The main presentation gives each supported family equal objective mass, so that the target reflects the average across disclosure families. We report both a stacked common-coefficient regression and an explicit average of separately estimated family coefficients. In the stacked regression, treatment variation also affects each family's coefficient contribution. The Online Appendix retains the market-cap alternatives, based on the baseline sample market value represented by adopting-country companies. These represent market value in the estimation sample, not a census of legally covered issuers. All weighting schemes, specifications and estimation outcomes remain in the coefficient registry.

The secondary specifications add lagged log income per capita, GDP growth and trade/GDP; use two-year windows; and examine alternative liquidity proxies. Event-time coefficients show differences before and after implementation. These checks assess the comparison, but cannot establish parallel trends or remove all coincident reforms (Roth et al. 2023, Rambachan and Roth 2023).

3.3 International implementation tests

We estimate implementation-hazard models with family and year effects, and linear probability models with country–family and family–year effects. The latter absorb persistent country-family differences and family-specific international waves. Independent national choices form the main comparison; coordinated implementations enter a separate comparison. All signals use the preceding year or earlier. These models explain documented implementation timing. Because the political decision can precede implementation, they do not establish that governments used a signal before making their original decision.

Portfolio exposure is the share of observed foreign portfolio positions in countries that have implemented the requirement. We require documented histories to cover at least half of the observed portfolio mass. The liquidity-success signal measures lower trading costs among the same companies in earlier adopting countries. A donor must have implemented by $t-2$; its observed change ends in $t-1$. This is an observed foreign outcome, not an estimated liquidity improvement caused by the foreign policy. We include exposure, success and their interaction together.

Following Koning et al. (2018), growth learning compares prior-year growth among earlier and later adopters in a geographic network with documented histories. Copying uses a larger geographic peer fixed with GDP in 2000, distinguishing recent and older implementation; language peers provide a second proxy. Competition uses export-pattern peers selected in 2000. Financial dependence adds aid to IMF credit and SDR allocations relative to GDP. It does not identify a lender's policy conditions. Missing histories, portfolio positions and outcomes remain missing.

Both family weights are applied to the same sample for each comparison. We require at least

15 countries and sufficient implementation and nonimplementation observations. Unstable or unsupported models remain in the full results registry. The finite core matrix contains 160 fits across channels, audiences, implementation types, estimators and weights. We report country-cluster inference and retain the Holm adjustment across all 256 attempted focal tests (Holm 1979). The channel estimates are exploratory.

4 Liquidity Findings

4.1 Liquidity around implementation

Table 2 and Figure 3 report the equal-family comparisons. The stacked coefficient is -0.190 ($p = 0.011$), corresponding to 17.3% lower trading costs. The explicit average of family coefficients is -0.177 ($p = 0.019$), corresponding to 16.2% lower trading costs. The targets differ because the stacked regression also reflects treatment variation within families. Market-cap alternatives are reported in Online Appendix Table OA6; they are smaller and less precise, with confidence intervals containing zero.

The family estimates show why the pooling target matters (Online Appendix Table OA1). Expanded auditor reporting includes 20 adopting countries and produces a coefficient close to zero. Public bank reporting and environmental and social reporting have negative but imprecise estimates. The climate-category liquidity estimate is supported by mandatory greenhouse-gas reporting in the UK (2013) and the revised Vietnamese annual-report form (FY2021). It should not be read as an estimate for all climate-risk or sustainability reporting requirements. The supply-chain comparison includes Australia (2019), France (2017), the UK (2016) and the US market observation based on California's 2012 rule. California is a subnational obligation with an eligibility perimeter, not a federal mandate covering every US issuer.

4.2 Interpretation and supporting comparisons

The country-condition specifications, two-year windows and other liquidity proxies are reported in the Online Appendix. The selected climate calendar contains seven jurisdictions with milestones during 2000–2025, but the primary liquidity comparison has only two adopting countries. New Zealand's 2023 milestone and the 2025 milestones in Australia, Hong Kong, Malaysia and Singapore lack three subsequent years in data ending in 2025. Among supply-chain entries, Canada (2024), Germany (2023) and Switzerland (2023) are also too recent; Norway (2022) lacks eligible later-adopter comparison countries under the current design. These restrictions explain the regression samples; they do not establish the absence of rules elsewhere. The shorter window admits IFRS as an additional supported family.

Event-time coefficients assess the evolution around the milestone (Online Appendix Figure OA1). Differences before implementation and sensitivity to timing, sample composition and family weights limit a causal interpretation. Concurrent requirements present a further issue: the UK 2013 auditor and greenhouse-gas rules share treated market observations. Country clustering addresses dependence in inference, but cannot separate the economic effects of a simultaneous bundle. We interpret the estimates as conditional country-market comparisons around the documented requirements.

Confidential tax reporting is reported separately in the Online Appendix. Its point estimates do not establish the same liquidity pattern as public disclosures. This comparison is informative about audience and reporting purpose, but does not isolate a tax-enforcement mechanism. It is not included in the public-disclosure headline average.

5 International Policy Diffusion

5.1 Diffusion Mechanism 1 – Learning from observed outcomes

Table 3 asks whether foreign-policy exposure has a different association with implementation when earlier adopters experience better liquidity. Under equal family weights, the exposure–success interaction is 0.832 ($p = 0.013$) in the hazard model. The comparison contains 202 country–family–years in 24 countries. Because continuous primitives are standardized, the interaction concerns their joint variation rather than a percentage-point effect on implementation probability. Online Appendix Table OA12 reports both family weights.

The positive hazard interaction is consistent with outcome-based learning. The within-country interaction is not positive, however, and the hazard interaction does not survive the full multiple-testing adjustment. Observed foreign liquidity can also reflect developments unrelated to the policy. The evidence therefore identifies an exploratory conditional association, rather than a demonstrated learning mechanism. Exposure to a foreign rule and observed success abroad convey different information, even when neither alone provides a stable explanation of implementation.

5.2 Diffusion Mechanism 2 – Copying larger or related economies

Online Appendix Table OA2 distinguishes following a larger economy from responding to its outcomes. The geographic peer is fixed before the adoption period. Recent and older implementations enter separately, following the timing distinction in Shipan and Volden (2008); language-peer adoption provides an additional familiarity measure. These comparisons do not show a consistent positive response across estimators and weights. Their coverage is narrower than the full policy calendar because the relevant peer's history must be observed.

The legal sources nevertheless identify concrete routes for international transmission. The Dutch standard setter's account of expanded reporting expressly refers to developments in the United Kingdom. National auditing requirements draw on international standards, while sustainability rules use international reporting frameworks. Such source evidence establishes the avail-

ability and sometimes use of a template. It does not by itself establish that the template produced a liquidity benefit or that a country copied it because it was successful.

5.3 Diffusion Mechanism 3 – Competition with economic rivals

The competition test uses prior implementation among countries with similar export patterns. The independent-choice comparison has limited support. In the comparison including coordinated implementations, the logit association is negative and survives the full focal-test adjustment. It does not show the predicted positive competitive response. The corresponding within-country comparison is different after absorbing persistent country-family differences and shared family-year waves. We therefore keep competition distinct from learning and from coordinated legal implementation.

5.4 Diffusion Mechanism 4 – External pressure and financial dependence

Following Koning et al. (2018), the financing test relates implementation to prior financial dependence. The within-country estimates do not provide a stable positive association under both weights. This proxy combines distinct financing relationships and does not identify policy conditionality. The legal classification of coordinated requirements records a separate institutional source of external influence. Joint-channel models are reported only where the intersection of the data meets the support requirements; the registry retains failures as well as fitted comparisons.

6 Conclusion

We bring public-disclosure requirements into a common country-level liquidity and implementation framework. Equal-family comparisons indicate lower trading costs following implementation; market-cap weighting produces smaller and less precise estimates. The differences across families, timing and comparison samples prevent a conclusion that every disclosure rule generates a common liquidity improvement.

The implementation tests distinguish the presence of foreign rules from the outcomes observed

among earlier adopters. A positive exposure–liquidity-success interaction appears in the hazard models, but complementary within-country models and the full multiple-testing adjustment do not confirm it. Together, the analyses provide a global source-linked basis for studying the market consequences and international transmission of disclosure requirements, while leaving causal attribution to individual rules and policymakers' learning unresolved.

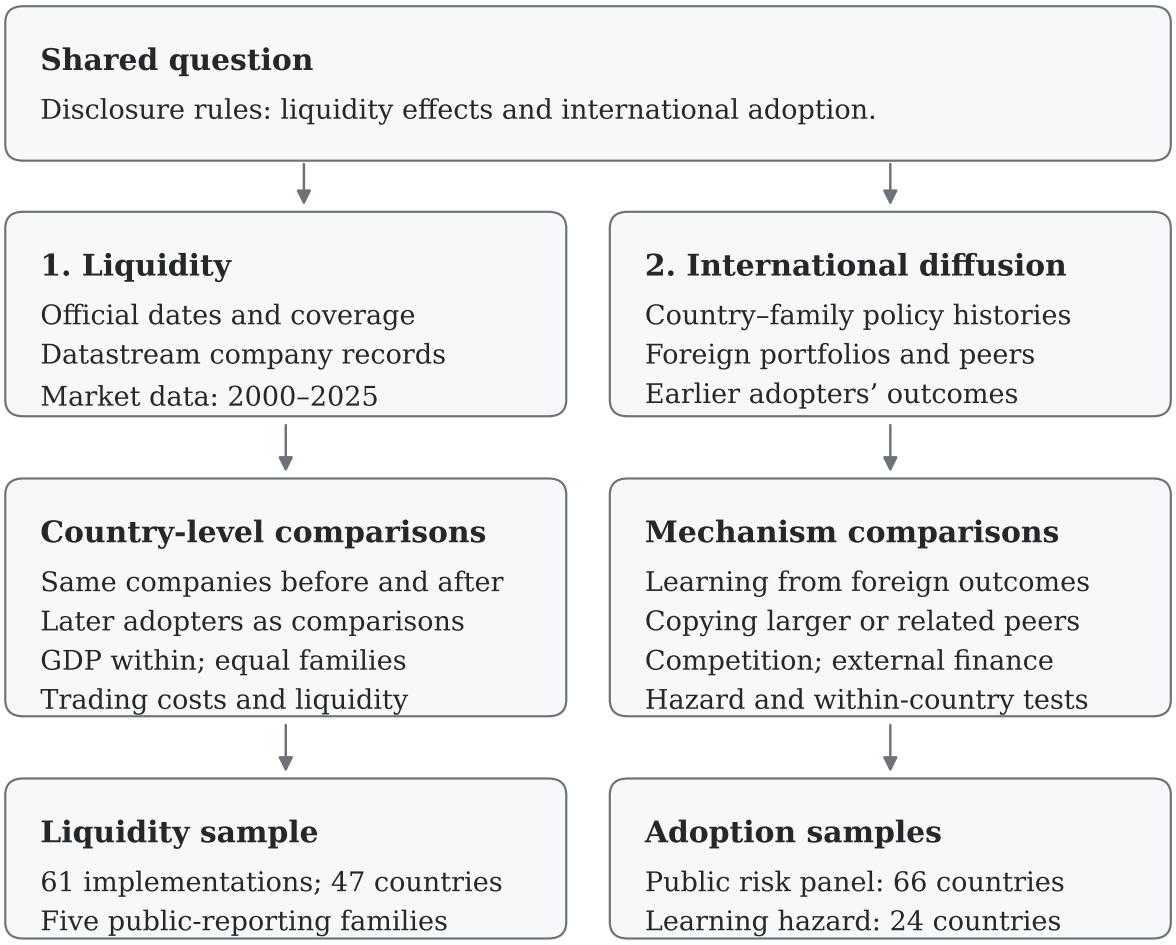


Figure 1. Disclosure regulation, liquidity and international diffusion
The framework separates the market comparison from the adoption-mechanism comparison. Samples differ between branches. Arrows describe the research design, not established causal effects. Alternative weights and supporting models remain in the Online Appendix.

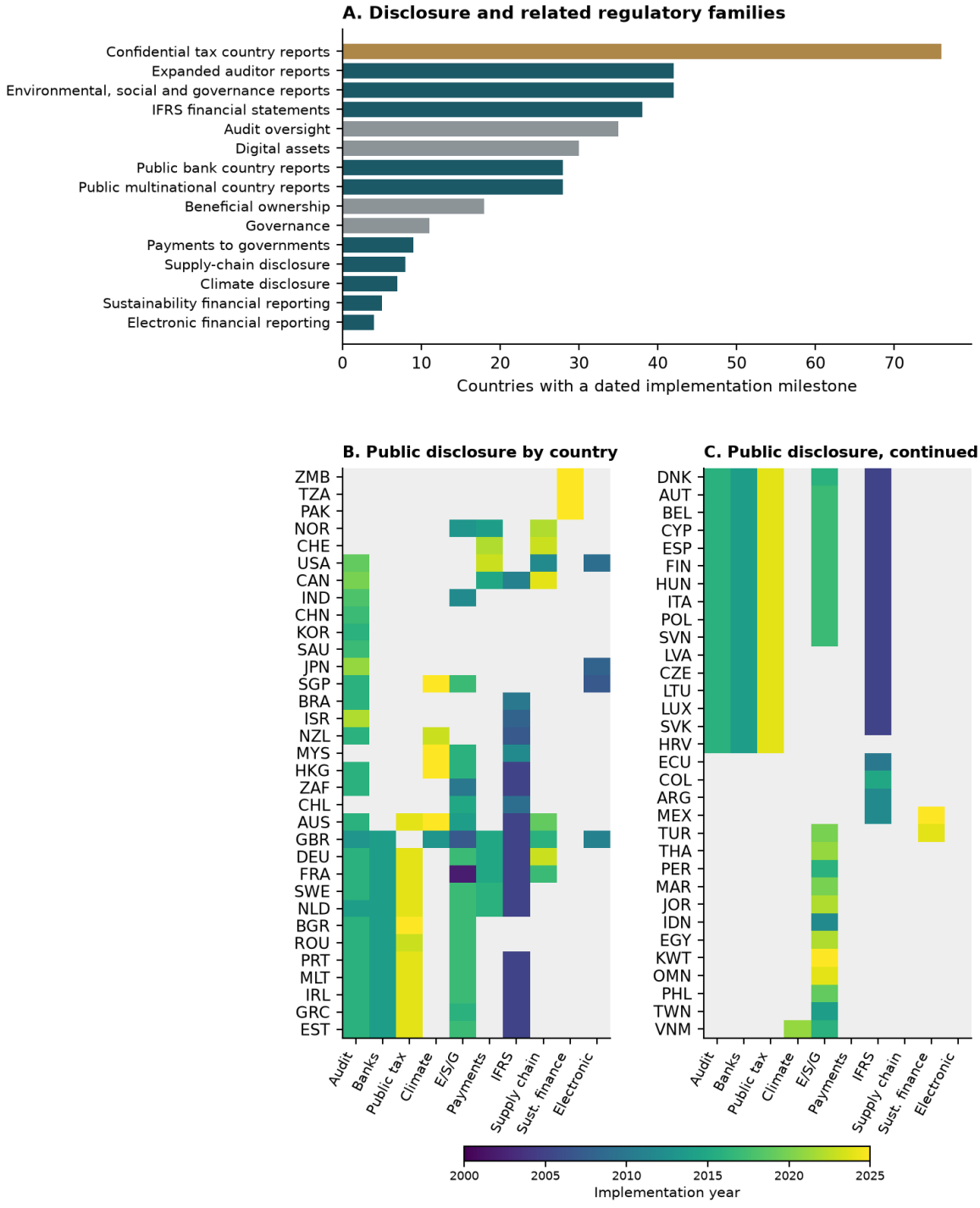


Figure 2. Policy families and international implementation

Panels show documented milestones dated 2000–2025 in the 90-jurisdiction universe. Country profiles are grouped by similarity in recorded public-disclosure families. Grey means no dated milestone in this calendar, not verified absence of reporting rules. This is not a complete global legal census; broader ESG rules can contain climate disclosures.

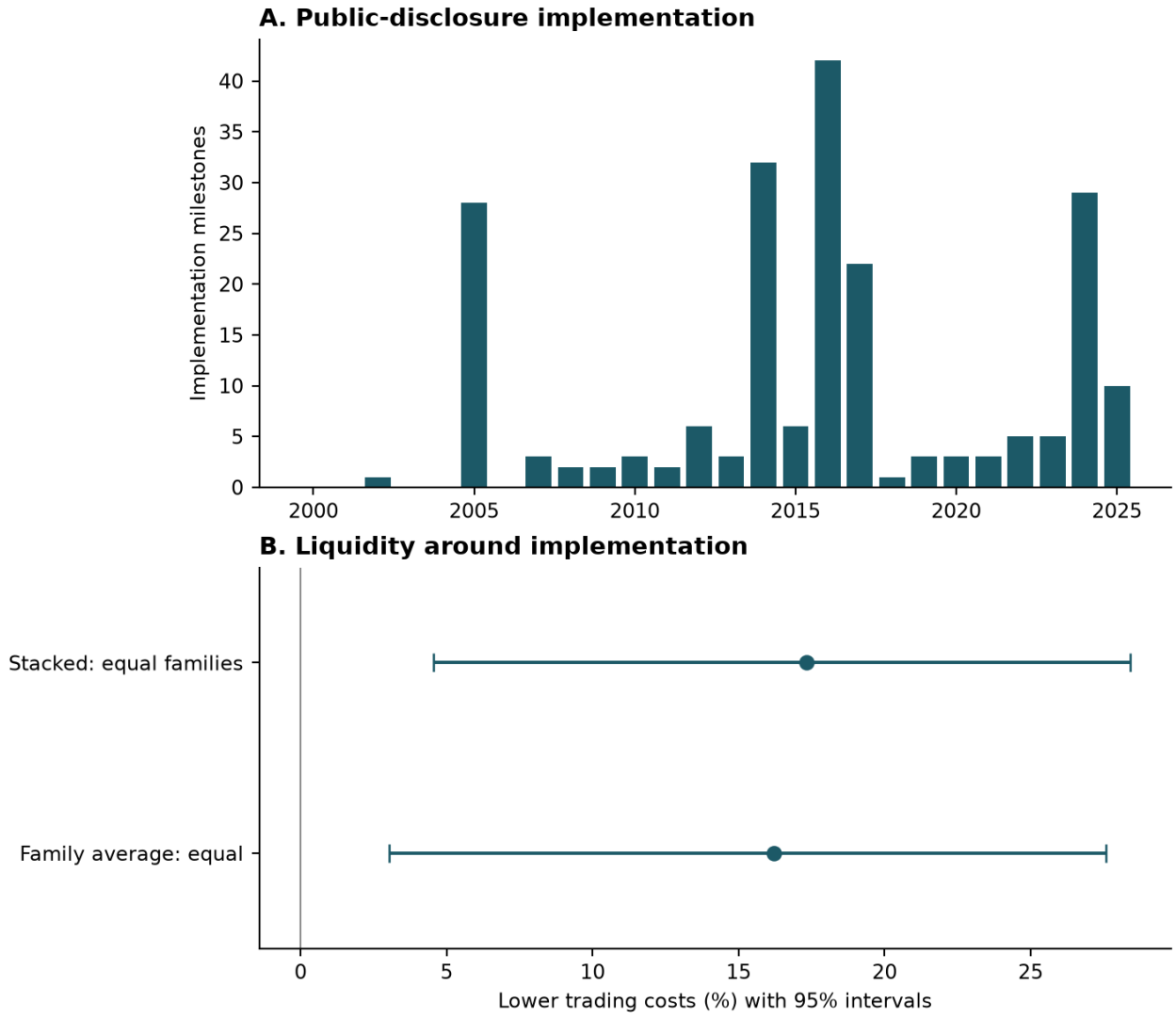


Figure 3. Public-disclosure implementation and stock liquidity

Panel A reports 211 country–family milestones. Panel B shows two equal-family estimates of lower trading costs, $100[1-\exp(\widehat{\beta})]$, with country-cluster 95% intervals. The supported liquidity sample contains 61 implementations and 47 countries. Positive values denote lower trading costs, not a proportionate increase in liquidity itself. Alternative family weights remain in Online Appendix Table OA6.

Table 1. Public-disclosure implementation milestones

Family	Countries	Liquidity adopters
Expanded auditor reports	42	20
Public bank country reports	28	13
Public multinational country reports	28	–
Climate disclosure	7	2
Environmental, social and governance reports	42	22
Payments to governments	9	–
IFRS financial statements	38	–
Supply-chain disclosure	8	4
Sustainability financial reporting	5	–
Electronic financial reporting	4	–

This table reports documented country–family implementation milestones dated 2000–2025, not counts of distinct laws. A country may appear in several families. The final column counts adopting countries in the three-year liquidity design. A dash means the design lacks a supported cohort, not that the rule has no effect. Only the initial milestone selected for each country–family series is tabulated; amendments, successor phases, voluntary guidance and later schedules remain in the source archive. The US supply-chain entry is California legislation, not a federal mandate. The ESG category includes mandated reports with different E, S and G content, materiality and disclosure-or-explanation provisions; it does not imply an unconditional mandate for every metric.

Table 2. Disclosure implementation and stock liquidity

	(1)	(2)
Family weights	Equal	Equal
Aggregation	Stacked	Average
Trading costs	-0.190^{**}	-0.177^{**}
	(0.071)	(0.073)
Trading-cost change	-17.3%	-16.2%
Country–stack–years	1404	1404
Countries	47	47
Families	5	5
Joint lead-test p	0.026	0.016

This table reports country-level rollout estimates. The outcome is log Bid-Ask Spread; lower values indicate lower trading costs and greater stock liquidity. Each cohort uses fixed companies in three pre-implementation years and years one to three after the documented milestone, excluding the implementation year. Country weights use GDP in the preceding year. Each supported family receives equal objective mass. Market-cap alternatives are reported in Online Appendix Table OA6. Their estimates are smaller and less precise. A stacked objective and an explicit average of family coefficients are distinct pooling targets. Standard errors in parentheses cluster by country and preserve dependence across policy stacks. *, ** and *** denote unadjusted significance at 10%,5% and 1%. These are country-policy comparisons, not verified effects on every legally covered issuer.

Table 3. Foreign policy exposure and observed liquidity outcomes

	(1)	(2)
Estimator	Logit	Within
Family weights	Equal	Equal
Foreign-policy exposure	0.372	0.084
	(0.554)	(0.050)
Successful foreign adopters	-1.851^{**}	0.002
	(0.740)	(0.044)
Exposure x successful adopters	0.832^{**}	-0.022
	(0.310)	(0.023)
Country–family–years	202	432
Countries	24	24
Implementations	25	26
Families	5	6

This table reports associations with documented implementation timing. Foreign-policy exposure is the lagged portfolio-weighted share of countries with the requirement. Successful foreign adopters are the share of observed earlier adopters with falling same-company log trading costs, measured before the focal implementation year; these are observed outcomes, not estimated effects caused by the foreign rule. Both primitives and their product enter together with lagged income, GDP growth and trade. Continuous primitives are standardized before the interaction. Logit includes family and year effects; Within denotes a linear probability model with country–family and family–year effects. Logit coefficients are log odds; within coefficients are probability changes. Country-cluster standard errors appear in parentheses. The samples differ because logit removes fixed-effect groups without outcome variation. Nominal stars use the conventions in Table 2; the full 160-model,256-focal-test adjustment is retained in the results files. Both family weights are reported in Online Appendix Table OA12. Decision dates are not available for all cases, so these models do not establish learning at the political decision date.

A Variable Definitions

Table A1. Variables, Sources and Regression Terms

Variable or term	Definition and construction
Cleaned daily returns	Retain returns between -1 and 2. Exclude an adjacent-return pair when either daily return exceeds one and the compounded two-return change is below 0.2. The retained price is finite and at least US$0.10; a country trading date requires at least five securities with positive volume and sufficient price.
Country–family milestone, g	Source-supported implementation year for a country and family. The date-basis field distinguishes legal force, covered period, audit engagement and filing obligation. Unknown dates are not untreated observations.
Reporting family and audience	Ten public categories in Table 1. Confidential tax CbCR remains separate. A category does not establish universal issuer eligibility or mandatory disclosure of every item.
Treated country, D_{cg}	One for the implementing country in its family–cohort stack; zero for an eligible documented later adopter whose milestone follows the outcome window. Require at least one adopting and three comparison countries per stack.
Post-implementation, $Post_{gt}$	One in years $g+1$ to $g+3$ and zero in years $g-3$ to $g-1$. Omit year g. The shorter comparison uses two years on either side.
Treatment interaction	Product $D_{cg}Post_{gt}$. Event-time regressors replace Post with relative-year indicators. Year -1 is the reference; year zero is omitted.
Fixed-company sample	Select each company's largest positive-market-value security line in $g-1$. Retain the same company identifier, country and all four annual outcomes throughout the window. Require at least 30 companies per country–cohort.
Daily relative trading cost	Datastream quotes give $2(Ask-Bid)/(Ask+Bid)$ when $Bid \geq 0$, $Ask \geq Bid$ and $Ask+Bid>0$. The annual measure is the median of eligible daily observations.
Log trading costs	Log of the positive annual median relative trading cost, with at least 120 eligible observations. Winsorize positive annual medians at the year-specific first and ninety-ninth percentiles before logging. Country outcomes average company logs.
Dollar volume	US-dollar closing price times volume, with positive volume and a product of at least US$100. Missing price or volume remains missing.
Log price impact	Log of 10^8 times the annual median absolute cleaned daily return divided by dollar volume. Require at least 120 eligible observations and more than one distinct price; winsorize positive annual medians as above before logging.
Zero-return days	Fraction of observed cleaned daily returns equal to zero, with at least 120 eligible returns and more than one distinct price. Missing returns are not zeros. Country outcomes average company fractions.
Illiquidity index	Equal average of standardized log trading costs, log price impact and zero-return fraction. Calibration uses positive-cap, one-line-per-company observations in 2000. All components must exist. Higher values indicate less liquidity.
Country GDP weights	Constant-US-dollar GDP in $g-1$ from the archived World Bank series, normalized within stack. Cohort mass within family reflects the GDP represented by adopting countries.
Equal family mass	Give each supported family equal normalized objective mass. Residual treatment variation also affects a stacked coefficient; an explicit average uses family masses directly.

Variables, Sources and Regression Terms (continued)

Variable or term	Definition and construction
Market-cap family mass	Alternative mass from baseline sample market value represented by adopting-country companies. It is not a census of every covered issuer or all global market value. Liquidity and diffusion use their stated supported cases.
Fixed effects	Liquidity absorbs country and year effects within cohort stack. Diffusion logit includes family and year effects. Within-country diffusion absorbs country–family and family–year effects.
First implementation event	One in the documented country–family implementation year; zero in its documented pre-event risk years. Later cases can provide right-censored pre-event observations through 2025. Unknown histories remain missing.
Independent national choice	The primary diffusion sample excludes observations classified as coordinated obligations. The all-documented comparison retains them. This is a legal classification, not an estimated policymaker motive.
Portfolio exposure, P	Sum of prior-year portfolio weights on partners implementing by $t-1$. Use positive observed IMF total-economy asset positions, exclude self positions and normalize over partners with documented family histories.
Known-history portfolio coverage	Positive portfolio mass in documented-history partners divided by all observed positive foreign portfolio mass. The core tests require at least 50%. Unreported bilateral positions are not imputed zeros.
Foreign liquidity improvement	Negative change in mean company log trading costs from $t-2$ to $t-1$, holding companies and mapping fixed with at least 30 pairs. Positive means lower observed costs, not an estimated policy-caused gain.
Successful-adopter share, S	Unweighted share of observed eligible foreign adopters with positive liquidity improvement. Partners must implement by $t-2$; outcome-observed partners must cover at least 80% of eligible-adopter portfolio mass. No eligible adopters gives zero signal; missing required outcomes remain missing.
Successful-adopter exposure, *Lexp*	Sum of portfolio weights on eligible earlier adopters with positive observed liquidity improvement. This weighted exposure differs from the unweighted share S.
Learning interaction, $P \times S$	Product of standardized exposure and success primitives, with both lower-order terms included. The logit coefficient is a log-odds interaction, not a percentage-point probability effect.
Larger peer, B	Whether the first larger-GDP peer among fixed geographic neighbours implemented by $t-1$. Fix size with GDP in 2000. An absent peer or undocumented history is not coded zero.
Recent and older copying	For that same larger peer, $Bnew = 1$ if implementation is in $t-1$; $Bold = 1$ if earlier than $t-1$. Retain the documented-history requirement.
Language-peer implementation	Prior-year adoption share among official-language peers with documented histories; require at least three observed peers. The proxy does not establish actual copying.
Trade-competitor implementation	Prior adoption share among export-pattern peers selected with the archived 2000 network. Require at least three documented histories; do not reselect peers based on current outcomes.
Growth learning	Indicator that median prior-year GDP growth among earlier adopters exceeds the median among later adopters in the ten nearest documented peers. Require at least 80% macro coverage in each group. No earlier adopters gives zero; if all peers adopted, compare their median with focal-country growth. Other missing support remains missing.
External financial dependence	Archived prior-year aid plus IMF credit and SDR allocations relative to GDP. The proxy does not identify actual financing conditions imposed by lenders.

Variables, Sources and Regression Terms (continued)

Variable or term	Definition and construction
Income per capita	Lagged log positive constant-price GDP per capita from the archived World Bank series; included in diffusion and controlled liquidity models.
GDP growth	Lagged annual GDP-growth percentage from the archived country series. Missing values are not replaced by assumed growth.
Trade openness	Lagged exports plus imports as a percentage of GDP from the archived World Bank series.
Standardized primitives	Within each diffusion sample, centre and scale nonbinary primitives and controls using population standard deviations. Binary variables retain their scale. Build interactions afterwards. Paired weights use identical samples and scaling.
Joint channel model	Exposure, successful-adopter exposure, larger-peer adoption, trade peers, language peers and financial dependence, with three country controls; estimated only on a supported data intersection.
Support and inference	Diffusion requires at least 50 rows, 15 countries, 10 events and 10 nonevents, suitable rank and variance. Logit drops fixed-effect groups without event variation. Standard errors cluster by country; nominal inference uses $G-1$ degrees of freedom.
Lead test and adjusted p-values	Liquidity uses a joint country-cluster Wald test of pre-event coefficients. Diffusion Holm adjustment covers 256 focal attempts in 160 fits; unestimated attempts receive $p=1$ for adjustment only.
Restricted five-model benchmark	Historical 30-implementation estimates retain stacked equal/cap mass, explicit family averages and covariance-weighted GLS. They are distinct historical benchmarks, not estimates on the expanded sample.

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